Data-Backed Journal-Style Replication of *Data Analytics, Innovation, and Firm Productivity*

Replication Paper Project

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Abstract

This paper presents a methodology-specific local analogue of Data Analytics, Innovation, and Firm Productivity using only processed datasets and result files available in the workspace. The replication preserves the source paper's core empirical logic while making every local substitution explicit. The paper is therefore reported as methodology-specific local analogue rather than as a source-data replay. The current local batch contains 23,901 observations across 184 entities, and the batch headline estimate is -0.0024 for AI Score x Breadth of Recombination with p = 0.74. The executable analogue is designed to recover the paper's asymmetry claim: analytics-like capability should be more useful when innovation comes from broad recombination than when it depends on entirely new technological moves with limited data support. The contribution is a journal-style replication manuscript that separates direct source claims from executable local evidence.

Keywords: data analytics; AI; innovation; recombination; inventive impact; local calculated analogue

1 Introduction

This paper begins from a paper-first reading of Data Analytics, Innovation, and Firm Productivity and treats the source article as an empirical argument that has to be rebuilt before it can be

judged. The present manuscript is classified as methodology-specific local analogue. A paper-first restatement of the empirical question is as follows: The source paper asks when data analytics capabilities are most valuable for innovation and whether they are especially complementary to innovation based on combining diverse existing knowledge (Wu et al., 2020; March, 1991; Levitt and March, 1988; Levinthal and March, 1993). The replication matters because claims about analytics and AI often become vague unless they are tied to specific innovation modes. The original article does not rest on a descriptive regularity alone. It ties a theory about search, differentiation, timing, or organizational learning to a specific unit of analysis, sample frame, and estimand. A careful replication therefore has to recover the structure of that logic before any local coefficients are interpreted (Wu et al., 2020; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001). The executable evidence base used in this manuscript contains 23,901 observations across 184 entities and spans 1986 to 2016 in the local executable panel. Those numbers are not filler. They define the actual inferential scope of the local reconstruction and set the boundary on what can be claimed honestly from the available datasets (Wu et al., 2020; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996; Jaffe, 1986; Hall et al., 2005). The paper's contribution is a transparent patent-level analogue that shows whether analytics-coded capability is more complementary to diverse recombination than to entirely new technological moves in the current local dataset. The paper is written to read like a formal journal article rather than an internal workflow memo. The main text develops the theoretical logic, data reconstruction, derivations, empirical design, and results in continuous prose, while the appendices preserve the audit trail required for reruns and later extensions (Wu et al., 2020; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996; Jaffe, 1986; Hall et al., 2005). The paper belongs to a top-journal literature on digital capabilities, innovation complementarity, and recombination. (Wu et al., 2020; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Cohen and Levinthal, 1990; Kogut and Zander, 1992, 1996; Teece et al., 1997; Benner and Tushman, 2003). That literature matters because the local rerun is not simply asking whether AI-coded patents are better. It is asking whether analytics-like capability changes the return to different innovation modes. (Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Henderson and Clark, 1990; Anderson and Tushman, 1986; Hargadon and Sutton, 1997; Tripsas and Gavetti, 2000; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996; Stuart, 2000). The present

manuscript therefore treats the source paper as one contribution inside a broader top-journal stream on exploration, recombination, performance measurement, alliance-based learning, inventive search, technological discontinuity, and patent-based inference. The literature review is intentionally broad because each replication paper should stand on a serious journal-level reference base rather than on the source article alone. (Rosenkopf and Almeida, 2003; Schilling and Phelps, 2007; Almeida and Kogut, 1999; Fleming et al., 2007; Jaffe, 1986; Jaffe et al., 1993; Trajtenberg, 1990; Hall et al., 2005; Henderson and Cockburn, 1994). The review therefore anchors the data-analytics paper inside the broader strategy literature on recombination and inventive performance. (Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002). A second topic-specific literature block is added to sharpen the manuscript's empirical boundary and identification problem. (Ghosh et al., 2014; Fleming and Sorenson, 2004; Hall et al., 2005). A final topic-specific literature block keeps the bibliography tied to the paper's actual mechanism rather than to a generic replication template. (Lavie et al., 2011; Rivkin and Siggelkow, 2003; Nerkar, 2003).

2 Original Study, Theory, and Replication Boundary

The focal source article is (Wu et al., 2020). The original study can be summarized as a mechanism paper with a sharp empirical core. The published paper argues that data analytics are more valuable for some kinds of innovation than for others. Diverse recombination is a central positive complement in the source argument. Entirely new technology creation is comparatively less compatible with data analytics in the published theory. That combination matters because the source article asks readers to believe not merely that two variables move together, but that a particular organizational or strategic mechanism explains why the relationship should appear in the data (Wu et al., 2020; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001). In a setting like this, faithful replication requires more than reusing a title and rerunning a regression. The theoretical contribution lives in the way constructs are defined, how they enter the estimand, and which comparison the model is supposed to adjudicate. For that reason, this manuscript first reconstructs the published theory and only then turns to the local executable analogue (Wu et al.,

2020; March, 1991; Levitt and March, 1988; Levinthal and March, 1993). The replication boundary is therefore explicit. Direct source facts are carried over when they can be read from the published article itself. Local replication decisions appear only when the current workspace must translate the source design into verified processed datasets. The goal is disciplined mechanism recovery, not rhetorical imitation that quietly changes the design (Wu et al., 2020; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996).

2.1 Original Paper Methodology

A faithful evaluation of Wu, Hitt, and Lou (2020) starts with the published empirical design rather than with the local rerun. The source article uses two linked empirical designs: a firm survey for process-improvement and new-technology-development outcomes and a larger patent analysis for public firms to evaluate analytics complementarity across innovation modes (Wu et al., 2020; March, 1991; Levitt and March, 1988; Levinthal and March, 1993). At the level of sample construction, the source article works from The published analysis combines a 331-firm survey with a broader patent sample of more than 2,000 publicly traded firms. At the level of measurement, it relies on The published design uses direct firm-level evidence on data analytics capabilities and relates that capability to process improvement, diverse recombination, and entirely new technology creation. These details matter because they determine what the published paper actually observes rather than what a later analogue can observe (Wu et al., 2020; March, 1991; Levitt and March, 1988; Levinthal and March, 1993). At the level of estimation, the published paper uses survey and patent-data models that test whether analytics capability is more complementary to process improvement and diverse recombination than to entirely new technologies. The article's central comparison is across innovation modes, asking whether analytics capability is disproportionately valuable when firms recombine a diverse set of existing technologies rather than generate entirely new technologies. Reading the source methodology this way prevents the current manuscript from quietly replacing the original design with one generic fixed-effects story (Wu et al., 2020; March, 1991; Levitt and March, 1988; Levinthal and March, 1993).

DesignElement	SourcePaperDesign
Published empirical setting	The published analysis combines a 331-firm survey with a broader patent sample of more than 2,000 publicly traded firms.
Published measurement architecture	The published design uses direct firm-level evidence on data analytics capabilities and relates that capability to process improvement, diverse recombination, and entirely new technology creation.
Published estimator	survey and patent-data models that test whether analytics capability is more complementary to process improvement and diverse recombination than to entirely new technologies
Published identification and comparison rule	The article's central comparison is across innovation modes, asking whether analytics capability is disproportionately valuable when firms recombine a diverse set of existing technologies rather than generate entirely new technologies.

Published empirical design reconstructed from the source paper before local estimation begins.

2.2 Replication Status and Boundary

Replication status for this paper: Methodology-specific local analogue (Wu et al., 2020; March, 1991; Levitt and March, 1988; Levinthal and March, 1993). This paper is a methodology-specific

local analogue because the original article directly measures analytics capability with firm-level survey and related data, whereas the local rerun uses patent-level AI classification scores. The present paper can test the complementarity logic inside the current patent panel, but it does not replay the original survey architecture (Wu et al., 2020; March, 1991; Levitt and March, 1988; Levinthal and March, 1993).

2.3 Direct Source Facts Versus Replication Decisions

The manuscript distinguishes direct source facts from local execution decisions. The source article establishes that The original paper is about complementarity between analytics capability and innovation mode. The local reconstruction therefore adopts the following bounded decision: AI classification score is used as the local analytics-capability analogue. This bridge is treated as a visible design choice, not as a hidden substitution. The reason for spelling out this bridge in prose is that replication credibility depends on showing exactly where the manuscript is following the article and where it is making an explicit local translation. That is especially important in this project, because the current paper is built from processed datasets already present in the workspace rather than from a reconstructed raw archival pipeline (Wu et al., 2020; March, 1991; Levitt and March, 1988; Levinthal and March, 1993). The manuscript distinguishes direct source facts from local execution decisions. The source article establishes that Diverse recombination is a positive complement in the published theory. The local reconstruction therefore adopts the following bounded decision: Breadth of recombination carries the diverse-recombination construct in the executable panel. This bridge is treated as a visible design choice, not as a hidden substitution. The reason for spelling out this bridge in prose is that replication credibility depends on showing exactly where the manuscript is following the article and where it is making an explicit local translation. That is especially important in this project, because the current paper is built from processed datasets already present in the workspace rather than from a reconstructed raw archival pipeline (Wu et al., 2020; March, 1991; Levitt and March, 1988; Levinthal and March, 1993). The manuscript distinguishes direct source facts from local execution decisions. The source article establishes that Entirely new technology is comparatively less compatible with analytics in the source paper. The local reconstruction therefore adopts the following bounded decision: Zero-prior-art patents provide the local entirely-new-technology analogue. This bridge is treated as a visible design choice, not as a hidden substitution. The reason for spelling out this bridge in prose is that replication credibility depends on showing exactly where the manuscript is following the

article and where it is making an explicit local translation. That is especially important in this project, because the current paper is built from processed datasets already present in the workspace rather than from a reconstructed raw archival pipeline (Wu et al., 2020; March, 1991; Levitt and March, 1988; Levinthal and March, 1993).

DirectSourceFact	LocalReplicationDecision
The original paper is about complementarity between analytics capability and innovation mode.	AI classification score is used as the local analytics-capability analogue.
Diverse recombination is a positive complement in the published theory.	Breadth of recombination carries the diverse-recombination construct in the executable panel.
Entirely new technology is comparatively less compatible with analytics in the source paper.	Zero-prior-art patents provide the local entirely-new-technology analogue.

Direct source facts and explicit local execution decisions carried in the body of the paper.

3 Data, Sample, and Measures

The local evidence chain is deliberately modular. The source paper is treated as the authority for the research question, theoretical mechanism, and published design. Estimation, by contrast, proceeds only through the verified local files. The required dataset types in this paper are Patent-level AI classification file already produced in the local regression output folder. Patent-level recombination and impact variables from the cleaned reproduction dataset. Firm-year science exposure already merged into the patent panel (Wu et al., 2020; Jaffe, 1986; Hall et al., 2005). Operationally, the reconstruction relies on four evidence layers: the analysis panel, the coefficient export, the sample summary, and, when available, the fit summary. The resulting executable panel supplies 23,901 observations across 184 entities and covers 1986 to 2016 in the local executable panel. Every numeric claim in the results section is intended to trace back to one of those verified evidence layers rather than to a hand-copied number from the article (Wu et al., 2020; Jaffe, 1986; Hall et al., 2005; Ahuja,

2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996). This data-lineage discipline matters for two reasons. First, it prevents the manuscript from smuggling unverified preprocessing steps into the design. Second, it forces every substitution to be visible. If a construct cannot be reproduced from the current processed datasets, the paper states the analogue directly instead of pretending that the original archive has been fully recreated (Wu et al., 2020; Jaffe, 1986; Hall et al., 2005; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001). The local analogue uses patent-level AI classification scores as a processed measure of analytics capability and tests whether those scores are more valuable in patents that recombine a broader range of existing technologies than in patents that look entirely new. The local sample frame is Patent-level recombination panel merged to AI classification scores and firm-year science exposure The observed time window in the executable panel is 1986 to 2016 in the local executable panel The outcome is Patent-level inventive impact (Wu et al., 2020; Jaffe, 1986; Hall et al., 2005). The main text works from a verified analysis panel, a verified coefficient export, and a verified sample summary. Exact file names are retained in the replication notes and appendices, while the article body focuses on data lineage and empirical function (Wu et al., 2020; Hall et al., 2005).

3.1 Exact Local Datasets Used in This Replication

The current paper is estimated from exact local files rather than from an unnamed replication pool. The executable evidence chain consists of ‘wu_2020_analysis_dataset.csv‘, ‘wu_2020_model_coefficients.csv‘, ‘wu_2020_sample_summary.csv‘, ‘wu_2020_model_fit.csv‘. Naming these files in the body matters because the paper should make clear what was actually used for estimation (Wu et al., 2020; Jaffe, 1986; Hall et al., 2005). The file-level roles are also specific rather than interchangeable: Estimate the local model and regenerate descriptive statistics is supplied by ‘wu_2020_analysis_dataset.csv‘; Recover the reported local coefficients and p-values is supplied by ‘wu_2020_model_coefficients.csv‘; Audit sample construction and raw grouped moments is supplied by ‘wu_2020_sample_summary.csv‘; Check model diagnostics when a fit file is available is supplied by ‘wu_2020_model_fit.csv‘. Together, these files support 23,901 observations across 184 entities over 1986 to 2016 in the local executable panel (Wu et al., 2020; Jaffe, 1986; Hall et al., 2005). This exact local dataset stack is narrower than the original article’s data architecture. Patent-level AI classification file already produced in the local regression output folder. Patent-level recombination and impact variables from the cleaned reproduction dataset. Firm-year science exposure already

merged into the patent panel. The manuscript therefore keeps the boundary visible: the source paper supplies the original methodology, whereas the listed local files supply the estimable analogue (Wu et al., 2020; Jaffe, 1986; Hall et al., 2005).

EvidenceLayer	DatasetFile	RoleInPaper
Analysis panel	wu_2020_analysis_dataset.csv	Estimate the local model and regenerate descriptive statistics
Coefficient export	wu_2020_model_coefficients.csv	Recover the reported local coefficients and p-values
Sample summary	wu_2020_sample_summary.csv	Audit sample construction and raw grouped moments
Fit summary	wu_2020_model_fit.csv	Check model diagnostics when a fit file is available

Exact local datasets and result files used in the present paper.

3.2 Executable File Selection and Data Availability

The present manuscript works from the processed datasets and calculated exports already available in the workspace for Wu, Hitt, and Lou (2020). That does not mean the paper skips data analysis. It means the data analysis begins from the current executable panel, current coefficient export, and current sample-summary files rather than from a freshly rebuilt raw archive. The substantive obligation is therefore to explain exactly why each verified file is needed and how it enters the paper's argument (Wu et al., 2020; Jaffe, 1986; Hall et al., 2005). The analysis panel used in this paper serves a specific empirical role: it is needed to estimate the local model and regenerate descriptive statistics. This evidence layer is not listed merely for bookkeeping. It is the concrete object that allows the manuscript to calculate the sample moments, estimators, and interpretation discussed in the main text. The body of the paper keeps the emphasis on evidence and function

rather than on machine-specific file labels (Wu et al., 2020; Jaffe, 1986; Hall et al., 2005; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996). The coefficient export used in this paper serves a specific empirical role: it is needed to recover the reported local coefficients and p-values. This evidence layer is not listed merely for bookkeeping. It is the concrete object that allows the manuscript to calculate the sample moments, estimators, and interpretation discussed in the main text. The body of the paper keeps the emphasis on evidence and function rather than on machine-specific file labels (Wu et al., 2020; Jaffe, 1986; Hall et al., 2005; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001). The sample summary used in this paper serves a specific empirical role: it is needed to audit sample construction and raw grouped moments. This evidence layer is not listed merely for bookkeeping. It is the concrete object that allows the manuscript to calculate the sample moments, estimators, and interpretation discussed in the main text. The body of the paper keeps the emphasis on evidence and function rather than on machine-specific file labels (Wu et al., 2020; Jaffe, 1986; Hall et al., 2005; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001). The fit summary used in this paper serves a specific empirical role: it is needed to check model diagnostics when a fit file is available. This evidence layer is not listed merely for bookkeeping. It is the concrete object that allows the manuscript to calculate the sample moments, estimators, and interpretation discussed in the main text. The body of the paper keeps the emphasis on evidence and function rather than on machine-specific file labels (Wu et al., 2020; Jaffe, 1986; Hall et al., 2005; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001). Taken together, these files form the minimum executable evidence chain behind the paper. The theory comes from the source article, the estimable sample comes from the processed analysis panel, the regression narrative comes from the current coefficient export, and the sample audit comes from the local summary file. That is the basis on which the manuscript makes every empirical claim (Wu et al., 2020; Jaffe, 1986; Hall et al., 2005; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001).

3.3 Sample Reconstruction

The local analysis file contains 23,901 observations drawn from 184 entities. The observed time span runs from 1986 to 2016. Selected sample moments are Impact = 5.309, Ai Score = 0.136, Diverse Recombination = 4.291 (Wu et al., 2020; Jaffe, 1986; Hall et al., 2005). These sample moments matter because they anchor the interpretation of the replication design. They indicate not only the scale of the local panel but also the baseline level of the key outcome and explanatory variables before the formal estimating equation is imposed (Wu et al., 2020; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996).

Statistic	Value
Observations	23,901
Unique Firms	184
Start Year	1986
End Year	2016
Mean Impact	5.309
Share Ai	8.74%
Mean Ai Score	0.136
Mean Diverse Recombination	4.291
Share Entirely New	0.00%

Sample reconstruction from the current local dataset.

3.4 Variable Construction and Derivation

The construct Data analytics capability is operationalized locally as AI classification score and appears in the executable dataset as ai_score_max_8. The derivation is not inferred loosely from the source article; it follows the verified local pipeline: Use the processed AI maximum score from the local patent classification output.. This mapping is retained because it preserves the construct's functional role in the original empirical argument while keeping the measurement auditable from the processed files used in estimation. The practical importance of this choice is that the manuscript can show how the concept enters the current dataset, how it is scaled, and how it interacts with

the rest of the specification instead of treating the variable as a black box. In replication terms, this is the point where conceptual comparability and executable traceability have to be balanced carefully, and the chosen local definition is intended to preserve that balance rather than maximize superficial similarity (Wu et al., 2020; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996; Jaffe, 1986; Hall et al., 2005). The construct Innovation by diverse recombination is operationalized locally as Breadth of recombination and appears in the executable dataset as diverse_recombination. The derivation is not inferred loosely from the source article; it follows the verified local pipeline: Carry forward breadth of recombination from the focal-patent panel.. This mapping is retained because it preserves the construct's functional role in the original empirical argument while keeping the measurement auditable from the processed files used in estimation. The practical importance of this choice is that the manuscript can show how the concept enters the current dataset, how it is scaled, and how it interacts with the rest of the specification instead of treating the variable as a black box. In replication terms, this is the point where conceptual comparability and executable traceability have to be balanced carefully, and the chosen local definition is intended to preserve that balance rather than maximize superficial similarity (Wu et al., 2020; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996; Jaffe, 1986; Hall et al., 2005). The construct Entirely new technology is operationalized locally as Zero-prior-art indicator and appears in the executable dataset as entirely_new_technology. The derivation is not inferred loosely from the source article; it follows the verified local pipeline: Classify patents with zero prior-art citations as entirely new technology cases.. This mapping is retained because it preserves the construct's functional role in the original empirical argument while keeping the measurement auditable from the processed files used in estimation. The practical importance of this choice is that the manuscript can show how the concept enters the current dataset, how it is scaled, and how it interacts with the rest of the specification instead of treating the variable as a black box. In replication terms, this is the point where conceptual comparability and executable traceability have to be balanced carefully, and the chosen local definition is intended to preserve that balance rather than maximize superficial similarity (Wu et al., 2020; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996; Jaffe, 1986; Hall et al., 2005). The construct Innovation output is operationalized locally as Inventive impact and appears in the executable dataset as inventive_impact. The derivation is not inferred loosely from the source article; it follows the verified local pipeline: Use the local inventive-impact measure..

This mapping is retained because it preserves the construct's functional role in the original empirical argument while keeping the measurement auditable from the processed files used in estimation. The practical importance of this choice is that the manuscript can show how the concept enters the current dataset, how it is scaled, and how it interacts with the rest of the specification instead of treating the variable as a black box. In replication terms, this is the point where conceptual comparability and executable traceability have to be balanced carefully, and the chosen local definition is intended to preserve that balance rather than maximize superficial similarity (Wu et al., 2020; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996; Jaffe, 1986; Hall et al., 2005). The construct Patent controls is operationalized locally as Science signal, degree, prior stock, firm age and appears in the executable dataset as science_signal; recombination_degree; total_patents_same_year; total_patents_until_prev_log; firm_age. The derivation is not inferred loosely from the source article; it follows the verified local pipeline: Reuse the patent-level controls already present in the merged panel.. This mapping is retained because it preserves the construct's functional role in the original empirical argument while keeping the measurement auditable from the processed files used in estimation. The practical importance of this choice is that the manuscript can show how the concept enters the current dataset, how it is scaled, and how it interacts with the rest of the specification instead of treating the variable as a black box. In replication terms, this is the point where conceptual comparability and executable traceability have to be balanced carefully, and the chosen local definition is intended to preserve that balance rather than maximize superficial similarity (Wu et al., 2020; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996; Jaffe, 1986; Hall et al., 2005).

Construct	Local measure	Derivation
Data analytics capability	AI classification score	Use the processed AI maximum score from the local patent classification output.
Innovation by diverse recombination	Breadth of recombination	Carry forward breadth of recombination from the focal-patent panel.
Entirely new technology	Zero-prior-art indicator	Classify patents with zero prior-art citations as entirely new technology cases.
Innovation output	Inventive impact	Use the local inventive-impact measure.

Construct	Local measure	Derivation
Patent controls	Science signal, degree, prior stock, firm age	Reuse the patent-level controls already present in the merged panel.

Variable mapping and derivation logic in the main text.

3.5 Construct-Level Derivation Logic

The derivation appendix records how each major construct enters the executable estimand for Wu, Hitt, and Lou (2020). The point is not to rehearse notation for its own sake, but to show how the source paper's conceptual vocabulary is translated into verified columns in the local panel. For Data analytics capability, the manuscript uses AI classification score under the exact local implementation ai_score_max_8. The derivation step is: Use the processed AI maximum score from the local patent classification output.. This construct is retained because it occupies the same analytic position in the local design that the published paper assigns to the original concept. For Innovation by diverse recombination, the manuscript uses Breadth of recombination under the exact local implementation diverse_recombination. The derivation step is: Carry forward breadth of recombination from the focal-patent panel.. This construct is retained because it occupies the same analytic position in the local design that the published paper assigns to the original concept. For Entirely new technology, the manuscript uses Zero-prior-art indicator under the exact local implementation entirely_new_technology. The derivation step is: Classify patents with zero prior-art citations as entirely new technology cases.. This construct is retained because it occupies the same analytic position in the local design that the published paper assigns to the original concept. For Innovation output, the manuscript uses Inventive impact under the exact local implementation inventive_impact. The derivation step is: Use the local inventive-impact measure.. This construct is retained because it occupies the same analytic position in the local design that the published paper assigns to the original concept. For Patent controls, the manuscript uses Science signal, degree, prior stock, firm age under the exact local implementation science_signal; recombination_degree; total_patents_same_year; total_patents_until_prev_log; firm_age. The derivation step is: Reuse the

patent-level controls already present in the merged panel.. This construct is retained because it occupies the same analytic position in the local design that the published paper assigns to the original concept.

4 Descriptive Analysis of the Current Dataset

Because the original paper is about differential complementarity across innovation modes, the descriptive analysis starts by locating AI-coded patents within the distribution of recombination breadth and novelty. For Inventive impact, the mean is 5.309, the standard deviation is 17.53, and the interquartile range runs from 1.000 to 6.000; For AI patent indicator, the mean is 0.0874, the standard deviation is 0.282, and the interquartile range runs from 0.00000 to 0.00000; For AI score, the mean is 0.136, the standard deviation is 0.294, and the interquartile range runs from 0.0038 to 0.0419; For Breadth of recombination, the mean is 4.291, the standard deviation is 4.563, and the interquartile range runs from 2.000 to 5.000 . The descriptive objective is not merely to show how common AI-coded patents are. It is to establish whether AI intensity, broad recombination, and entirely new technology cases occupy materially different parts of the patent sample (Wu et al., 2020; Jaffe, 1986; Hall et al., 2005; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001). The time profile is read as the arrival of analytics-like capability in the patent record. If AI-coded patents appear only late or only in narrow slices of the sample, the complementarity evidence has to be interpreted with that support in mind. the value of patent observations is 646 in 1986 and 534 in 2016; the value of mean inventive impact is 4.015 in 1986 and 3.768 in 2016; the value of share of AI-coded patents is 1.86% in 1986 and 41.20% in 2016 . These temporal shifts determine whether the later regression is identifying the paper's mechanism in a stable panel or in one whose observable context changes meaningfully across the sample window (Wu et al., 2020; Jaffe, 1986; Hall et al., 2005; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001). The nonparametric support check is treated as an analytics gradient, asking whether inventive impact shifts across the observed support of AI score before interaction terms are introduced. Mean outcomes range from 4.894 to

6.183 across the support of the focal regressor, while the mean regressor itself ranges from 0.0017 to 0.632. That pattern is useful because it shows what the formal model is trying to account for in the current dataset (Wu et al., 2020; Jaffe, 1986; Hall et al., 2005; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001). The correlation discussion is framed as a complementarity screen. The paper needs AI score to vary alongside, but not collapse into, recombination breadth, novelty, and science exposure. That is especially important in these replications, where construct translation can otherwise make the specification look more deterministic than it really is (Wu et al., 2020; Jaffe, 1986; Hall et al., 2005; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001).

4.1 Data Quality, Coverage, and Support

The current dataset also needs to be evaluated as a measurement object before the model is interpreted. Inventive impact has missing share 0.00%; AI patent indicator has missing share 0.00%; AI score has missing share 0.00% . At the other end of the support, Inventive impact remains close to complete with missing share 0.00%; AI patent indicator remains close to complete with missing share 0.00%; AI score remains close to complete with missing share 0.00% . This matters because the credibility of the regression depends on whether the focal constructs survive sample cleaning with enough support to be interpreted substantively (Wu et al., 2020; Jaffe, 1986; Hall et al., 2005). Coverage across entities is also visible in the current panel rather than assumed. INTERNATIONAL BUSINESS MACHS COR contributes 3,200 observations; GENERAL ELECTRIC CO contributes 1,318 observations; EASTMAN KODAK CO contributes 1,017 observations; MINNESOTA MINING & MFG CO contributes 854 observations; DU PONT E I DE NEMOURS & CO contributes 830 observations . The purpose of reporting these concentrations is to show whether the estimated mechanism is being learned from a broad panel or from a narrow set of heavily observed entities (Wu et al., 2020; Jaffe, 1986; Hall et al., 2005). The support of the focal regressor is broad enough to permit a distributional reading in addition to the formal model. In the lowest observed quantile, the mean regressor is 0.0017 and the mean outcome is 5.435 across 4,781 observations. In the highest observed quantile, the corresponding values are 0.632, 6.183, and 4,780 observations. This shift matters because it shows that the regression is being estimated on economically different parts of the observed support rather than on nearly identical cells (Wu et al.,

2020; Jaffe, 1986; Hall et al., 2005). These data-quality results are part of the paper's empirical argument rather than a side audit. The replication matters because claims about analytics and AI often become vague unless they are tied to specific innovation modes. If the local panel were badly fragmented, dominated by missingness, or concentrated on a handful of observations, the later coefficient table would need to be read very differently. Bringing this evidence into the body of the article is therefore necessary for an honest replication (Wu et al., 2020; Jaffe, 1986; Hall et al., 2005).

Variable	Missing	MissingShare
Inventive impact	0	0.00%
AI patent indicator	0	0.00%
AI score	0	0.00%
Breadth of recombination	0	0.00%
Entirely new technology	0	0.00%
Science signal	0	0.00%
Recombination degree	0	0.00%
Same-year patents	0	0.00%

Missingness profile for variables analyzed in the body of the paper.

Variable	N	Mean	SD	P25	Median	P75	Min	Max
Inventive impact	23,901	5.309	17.53	1.000	2.000	6.000	0	812
AI patent indicator	23,901	0.0874	0.282	0.0000	0.0000	0.0000	0	1
AI score	23,901	0.136	0.294	0.0038	0.0096	0.0419	0	1
Breadth of recombination	23,901	4.291	4.563	2.000	3.000	5.000	2	117
Entirely new technology	23,901	0.0000	0.0000	0.0000	0.0000	0.0000	0	0
Science signal	23,901	6.772	2.130	5.729	7.149	8.163	0	10
Recombination degree	23,901	0.555	0.174	0.444	0.560	0.691	0	1
Same-year patents	23,901	727.1	1,302	109.0	311.0	652.0	1	7321

Main descriptive statistics discussed in the body of the paper.

Period	Patents	MeanImpact	AIShare	MeanAIScore	MeanDiverseRecombination
1986	646	4.015	1.86%	0.0560	2.865
1987	793	4.296	1.89%	0.0594	2.914
1988	1,071	4.469	2.43%	0.0532	3.135
1989	1,398	3.926	2.43%	0.0573	3.084
1990	1,360	4.507	2.28%	0.0590	3.117
1991	1,333	4.749	2.18%	0.0503	3.151
1992	1,309	5.007	2.52%	0.0572	3.329
1993	1,358	5.678	4.05%	0.0799	3.465
1994	1,337	6.272	5.53%	0.0908	3.572
1995	1,217	6.260	5.26%	0.0960	3.844
1996	659	6.727	5.92%	0.102	3.838
1997	646	6.014	6.66%	0.114	4.245
1998	356	6.295	2.25%	0.0578	3.924
1999	654	5.483	5.05%	0.0875	3.810
2000	636	5.541	4.09%	0.0875	4.022
2001	614	5.151	5.86%	0.102	4.414
2002	632	4.889	6.17%	0.108	4.384
2003	522	4.502	5.17%	0.104	4.418
2004	324	3.664	6.79%	0.112	4.809
2005	284	4.500	14.08%	0.196	5.486
2006	336	4.890	13.99%	0.198	5.265
2007	246	3.654	8.13%	0.153	5.268
2008	535	3.652	27.10%	0.357	4.512
2009	206	4.801	15.53%	0.223	6.005
2010	511	4.417	11.35%	0.165	6.648
2011	651	6.452	16.90%	0.232	6.275
2012	1,068	4.000	21.07%	0.290	6.519
2013	868	5.386	16.59%	0.247	6.730
2014	725	12.64	16.14%	0.240	6.989
2015	1,072	6.616	26.59%	0.343	6.174
2016	534	3.768	41.20%	0.511	4.629

Observed time patterns in the current dataset.

Quantile	N	MeanRegressor	MeanOutcome
Q1	4,781	0.0017	5.435
Q2	4,780	0.0047	4.894
Q3	4,780	0.0100	4.926
Q4	4,780	0.0333	5.109
Q5	4,780	0.632	6.183

Grouped summaries used to describe raw patterns in the current dataset.

Variable	Inventive impact	AI patent indicator	AI score	Breadth of recombination	Entirely new technology	Science signal	Recombination degree	Same-year patents
Inventive impact	1.000	0.026	0.027	0.364	NA	0.020	0.018	0
AI patent indicator	0.026	1.000	0.884	0.046	NA	0.041	0.066	0
AI score	0.027	0.884	1.000	0.055	NA	0.055	0.073	0
Breadth of recombination	0.364	0.046	0.055	1.000	NA	0.023	0.466	0
Entirely new technology	NA	NA	NA	NA	NA	NA	NA	NA
Science signal	0.020	0.041	0.055	0.023	NA	1.000	0.037	0

Compact correlation structure for the main variables.

5 Empirical Strategy and Identification

The estimand targeted in this manuscript is A patent-level firm and year fixed-effects model of inventive impact on AI score, recombination breadth, entirely new technology, and the interactions of AI score with each innovation mode. The role of the empirical strategy is therefore to recover the sign, magnitude, and precision of the coefficients that embody the source paper's central mechanism (Wu et al., 2020; March, 1991; Levitt and March, 1988; Levinthal and March, 1993). Identification remains observational. The discipline in the design comes from preserving the source paper's unit

of analysis, keeping the core construct mapping explicit, and holding the local exclusion rules fixed throughout the rerun. This does not convert the study into an experimental design, but it does make the inferential claim narrower and more honest (Wu et al., 2020; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001). The first set of replication decisions concerns scope. The unit of analysis is Patent, the sample frame is Patent-level recombination panel merged to AI classification scores and firm-year science exposure, and the empirical window is 1986 to 2016 in the local executable panel. These choices define what counts as a comparable observation in the local reconstruction and prevent the paper from drifting into a broader but theoretically mismatched sample (Wu et al., 2020; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001). The second set of decisions concerns measurement and specification. The outcome is Patent-level inventive impact, and the main explanatory design is Patent-level fixed-effects model relating inventive impact to AI score, recombination breadth, entirely new technology, and two AI interactions. This is where the replication decides whether a variable is close enough to the published construct to remain in the estimand without changing the paper's substantive question (Wu et al., 2020; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001). The third set of decisions concerns statistical discipline. Controls and fixed effects are handled as follows: Science signal, recombination degree, same-year patenting, prior stock, firm age, and firm and year fixed effects.. The crucial exclusion or substitution rule is: The local rerun keeps only patents present in both the AI-classification output and the cleaned recombination-impact panel.. In practice, this means the local rerun prioritizes a stable estimand and transparent sample discipline over broader coverage that would weaken construct comparability (Wu et al., 2020; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001). A final design choice concerns interpretation. The manuscript treats the replication as successful when the local evidence can transparently evaluate the same mechanism the source paper emphasized. For this paper, the status is methodology-specific local analogue. When the local data architecture differs, the paper states that boundary directly rather than implying a source-data replay (Wu et al., 2020; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001).

DecisionPoint	LocalSpecification
Research question	The source paper asks when data analytics capabilities are most valuable for innovation and whether they are especially complementary to innovation based on combining diverse existing knowledge.
Unit of analysis	Patent
Sample frame	Patent-level recombination panel merged to AI classification scores and firm-year science exposure.
Time window	1986 to 2016 in the local executable panel.
Outcome definition	Patent-level inventive impact.
Main explanatory design	Patent-level fixed-effects model relating inventive impact to AI score, recombination breadth, entirely new technology, and two AI interactions.
Controls and fixed effects	Science signal, recombination degree, same-year patenting, prior stock, firm age, and firm and year fixed effects.
Exclusion or substitution logic	The local rerun keeps only patents present in both the AI-classification output and the cleaned recombination-impact panel.

Replication decision register retained in the main text.

5.1 Why These Design Decisions Are Necessary

Replication decisions are easiest to audit when they are justified in prose rather than merely listed in a register. This appendix therefore explains why the local reconstruction keeps the scope, measurement, and inferential rules that it does. Scope decisions: the manuscript works at the level of Patent inside the sample frame Patent-level recombination panel merged to AI classification scores and firm-year science exposure over 1986 to 2016 in the local executable panel. These are not cosmetic constraints. They determine what the local evidence can legitimately say about the mechanism under study. Measurement decisions: the outcome is Patent-level inventive impact and the main explanatory design is Patent-level fixed-effects model relating inventive impact to AI score, recombination breadth, entirely new technology, and two AI interactions. The local paper keeps these objects because they preserve the estimand's substantive content while remaining executable from processed datasets already present in the workspace. Inference decisions: Science signal, recombination degree, same-year patenting, prior stock, firm age, and firm and year fixed effects. The main exclusion or substitution rule is The local rerun keeps only patents present in both the AI-classification output and the cleaned recombination-impact panel. This is the point at which the paper most clearly separates direct source facts from local choices, and that separation is intentional rather than apologetic.

$$\log(1 + \text{Impact}_{pft}) = \beta_1 \text{AIScore}_p + \beta_2 \text{Breadth}_p + \beta_3 \text{AIScore}_p \times \text{Breadth}_p + \beta_4 \text{EntirelyNew}_p + \beta_5 \text{AIScore}_p \times \text{Entir} \tag{1}$$

6 Results

6.1 Main Estimates

The central local estimate is -0.0024 on AI Score x Breadth of Recombination (p = 0.74). The executable analogue is designed to recover the paper's asymmetry claim: analytics-like capability should be more useful when innovation comes from broad recombination than when it depends on entirely new technological moves with limited data support. Recombination degree is estimated at -0.440 with p = ¡ .001. Breadth of recombination is estimated at 0.0419 with p = ¡ .001. AI

score is estimated at 0.210 with p = 0.003 (Wu et al., 2020; Hall et al., 2005). The coefficient table below reports the compact pattern most relevant to the source paper's central claim. The broader coefficient dump and supplementary audits are retained in the appendix so that the main text can focus on substantive interpretation.

Term	Estimate	Std. error	P-value
AI score	0.2098	0.0650	0.0030
Breadth of recombination	0.0419	0.0091	7.37e-05
Science signal	0.0018	0.0289	0.952
Recombination degree	-0.4400	0.0871	2.00e-05
Same-year patents	-4.48e-05	2.08e-05	0.0392
Log prior patent stock	-0.1230	0.0648	0.0675
AI score x breadth of recombination	-0.0024	0.0072	0.743

Key coefficient table used in the main text.

6.2 Detailed Interpretation of the Local Results

The coefficient pattern is not being summarized as a bare sign check. Recombination degree is estimated at -0.440 with a p-value of ¡ .001; Breadth of recombination is estimated at 0.0419 with a p-value of ¡ .001; AI score is estimated at 0.210 with a p-value of 0.003; Same-year patents is estimated at -4.48e-05 with a p-value of 0.039 (Wu et al., 2020; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Jaffe, 1986; Hall et al., 2005). Within the focal specification, the sharpest local signal comes from Recombination degree, which is estimated at -0.440 with p = ¡ .001. That term matters because it is the coefficient most closely tied to the source paper's substantive claim rather than to a generic control adjustment (Wu et al., 2020; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Jaffe, 1986; Hall et al., 2005). The scale of the underlying variables also matters for interpretation. The outcome Inventive impact has mean 5.309 and standard deviation 17.53 in the local panel, while the main regressor AI score has mean 0.136 and standard deviation 0.294

(Wu et al., 2020; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Jaffe, 1986; Hall et al., 2005). A simple grouped comparison also helps clarify the pattern. Across the regressor quantiles, the mean outcome ranges from 4.894 to 6.183 while the mean regressor ranges from 0.0017 to 0.632. This does not substitute for the model, but it shows how much raw variation the specification must explain (Wu et al., 2020; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Jaffe, 1986; Hall et al., 2005). Taken together, the descriptive evidence and formal estimates should be read as a structured assessment of the paper's mechanism. The executable analogue is designed to recover the paper's asymmetry claim: analytics-like capability should be more useful when innovation comes from broad recombination than when it depends on entirely new technological moves with limited data support. The substantive question is not whether every coefficient matches a published table entry digit for digit, but whether the local evidence supports the same directional and conceptual interpretation (Wu et al., 2020; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Jaffe, 1986; Hall et al., 2005).

6.3 Coefficient Families in the Current Estimation Output

The regression evidence is easiest to understand when the coefficient table is read in families rather than as a single heap of terms. The paper therefore separates the estimates by model and highlights the strongest positive, strongest negative, and most focal signals within each specification (Wu et al., 2020; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996). Within baseline, the leading coefficient family reads as follows: Recombination degree = -0.440 (p = ¡ .001); Breadth of recombination = 0.0419 (p = ¡ .001); AI score = 0.210 (p = 0.003); Same-year patents = -4.48e-05 (p = 0.039); Log prior patent stock = -0.123 (p = 0.067). The largest positive estimate in this specification is AI score at 0.210, while the largest negative estimate is Recombination degree at -0.440. Reading the specification this way matters because it shows which part of the model is doing the explanatory work and which part is mainly serving as adjustment structure (Wu et al., 2020; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996). This fuller coefficient reading is

especially important for Wu, Hitt, and Lou (2020) because the source paper's contribution is not exhausted by one headline estimate. The current dataset can confirm some parts of the published mechanism more strongly than others, and a formal replication paper should show that uneven pattern directly in the main text (Wu et al., 2020; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996).

6.4 Model Comparison Using the Current Dataset

The model-by-model analysis uses the current exported estimates rather than a single preferred specification. In Baseline, the most informative focal term is Recombination degree with estimate -0.440 and p = ¡ .001 (Wu et al., 2020; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001). The Baseline specification contributes its own reading to the replication. Recombination degree is estimated at -0.440 with p = ¡ .001 and is classified as a strong signal; Breadth of recombination is estimated at 0.0419 with p = ¡ .001 and is classified as a strong signal; AI score is estimated at 0.210 with p = 0.003 and is classified as a strong signal . This matters because the current dataset may preserve one part of the published mechanism more clearly than another, and the paper should show that pattern directly rather than compress it into a single headline coefficient (Wu et al., 2020; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001). This model comparison matters for Wu, Hitt, and Lou (2020) because the source article's claim can only be evaluated responsibly if the pattern persists across the outcome or specification variants that are actually present in the local results folder. Put differently, the current dataset should be allowed to show where the mechanism is stable, where it weakens, and where it changes sign under a different outcome definition (Wu et al., 2020; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001).

Term	Model	Estimate	PValue	Signal	Reading
AI score	Baseline	0.2098	0.003	Strong	Positive estimate with p = 0.003.

Term	Model	Estimate	PValue	Signal	Reading
Breadth of recombination	Baseline	0.0419	¡ .001	Strong	Positive estimate with p = ¡ .001.
Science signal	Baseline	0.0018	0.95	Weak	Positive estimate with p = 0.95.
Recombination degree	Baseline	-0.4400	¡ .001	Strong	Negative estimate with p = ¡ .001.
Same-year patents	Baseline	-4.48e-05	0.039	Moderate	Negative estimate with p = 0.039.
Log prior patent stock	Baseline	-0.1230	0.067	Suggestive	Negative estimate with p = 0.067.
AI score x breadth of recombination	Baseline	-0.0024	0.74	Weak	Negative estimate with p = 0.74.

Model-by-model interpretation table drawn from the current coefficient exports.

6.5 Diagnostic Evidence

Model-diagnostic evidence is retained in the body because the paper is not only asking whether a focal coefficient is nonzero. It is also asking whether the specification is being estimated on a sample and likelihood structure that are transparent enough to justify interpretation (Wu et al., 2020; Jaffe, 1986; Hall et al., 2005). For data_analytics_innovation, the current fit export reports N = 23,884, pseudo R-squared = 0.0514, R-squared = 0.133 (Wu et al., 2020; Jaffe, 1986; Hall et al., 2005). These diagnostics do not, by themselves, validate the source paper's theory. They do, however, show that the local data analysis behind Wu, Hitt, and Lou (2020) is being carried out on a clearly described estimating sample with observable fit statistics (Wu et al., 2020; Jaffe, 1986; Hall et al., 2005).

model	n	logLik	AIC	BIC	pseudo_r2	r2	within_r2
data_analytics_innovation	23,884	-31,472	63,352	65,000	0.0514	0.133	0.0348

Model fit statistics discussed in the main results section.

Term	Estimate	Std. error	Statistic	P-value
AI score	0.2098	0.0650	3.227	0.0030
Breadth of recombination	0.0419	0.0091	4.591	7.37e-05
Science signal	0.0018	0.0289	0.0606	0.952
Recombination degree	-0.4400	0.0871	-5.054	2.00e-05
Same-year patents	-4.48e-05	2.08e-05	-2.157	0.0392
Log prior patent stock	-0.1230	0.0648	-1.897	0.0675
AI score x breadth of recombination	-0.0024	0.0072	-0.331	0.743

Detailed coefficient table from the current local estimation output.

7 Comparison to the Published Study

7.1 Original Results Versus the Current Local Dataset

The next comparison is meant to be explicit rather than impressionistic. The original paper's central empirical claim, the main measurement layer, and the effective sample definition are set alongside the current local dataset result so the reader can see where agreement is substantive and where it is conditional on different data architecture. (Wu et al., 2020). For Wu, Hitt, and Lou (2020), the local headline result is AI Score x Breadth of Recombination = -0.0024 with p = 0.74. The value of presenting that estimate directly against the original paper's reported finding is that it shows whether the mechanism survives a different but transparent processed dataset rather than asking the reader to infer comparability from prose alone.

Aspect	OriginalPaper	LocalDataset	Reading
Main empirical result	The published paper finds that data analytics are more valuable for process-oriented and diverse-recombination innovation than for entirely new technologies.	AI Score x Breadth of Recombination = -0.0024 with p = 0.74.	The published paper argues that data analytics are most useful when firms innovate by recombining diverse existing knowledge and less useful for entirely new technologies. The local analogue translates that claim into patent-level AI scores, recombination breadth, and an entirely-new-technology indicator.
Measurement and data construction	The published design uses direct firm-level evidence on data analytics capabilities and relates that capability to process improvement, diverse recombination, and entirely new technology creation.	The local analogue uses patent-level AI classification scores as a processed measure of analytics capability and tests whether those scores are more valuable in patents that recombine a broader range of existing technologies than in patents that look entirely new.	This row isolates how the executable local dataset differs from the original measurement architecture while preserving the same substantive target.

Aspect	OriginalPaper	LocalDataset	Reading
Sample and unit of analysis	The published analysis combines a 331-firm survey with a broader patent sample of more than 2,000 publicly traded firms.	Patent-level recombination panel merged to AI classification scores and firm-year science exposure. Local sample = 23,901 observations across 184 entities.	This row clarifies whether agreement or disagreement with the original paper could reflect sample translation rather than theory alone.

Comparison of the original paper's results and design to the current local dataset analysis.

7.2 Interpretive Implications of the Comparison

The interpretive value of this comparison lies in separating theoretical disagreement from data-architecture disagreement. The published study relies on the following measurement layer: The published design uses direct firm-level evidence on data analytics capabilities and relates that capability to process improvement, diverse recombination, and entirely new technology creation By contrast, the local paper relies on the following executable design: The local analogue uses patent-level AI classification scores as a processed measure of analytics capability and tests whether those scores are more valuable in patents that recombine a broader range of existing technologies than in patents that look entirely new These are not trivial implementation details. They determine whether the focal construct is being observed through text semantics, event timing, archival search coding, or a processed analogue built from the current workspace (Wu et al., 2020; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Jaffe, 1986; Hall et al., 2005). The same caution applies to the sample itself. The published design is described as follows: The published analysis combines a 331-firm survey with a broader patent sample of more than 2,000 publicly traded firms The local sample instead works from Patent-level recombination panel merged to AI classification scores and firm-year science exposure and yields 23,901 observations across 184

entities. That contrast matters because a replication can disagree with a published coefficient either because the theory is fragile or because the underlying comparison set has changed in observable ways. A careful paper should keep those possibilities analytically separate (Wu et al., 2020; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Jaffe, 1986; Hall et al., 2005). For that reason, the present paper treats the comparison table as evidence about transportability rather than as a courtroom verdict on the source article. The published paper argues that data analytics are most useful when firms innovate by recombining diverse existing knowledge and less useful for entirely new technologies. The local analogue translates that claim into patent-level AI scores, recombination breadth, and an entirely-new-technology indicator. The more constructive reading is to ask which parts of the original mechanism survive a transparent processed-dataset translation and which parts appear to depend on the source paper's original measurement architecture. That reading makes the replication useful both for evaluating the original claim and for guiding the next round of executable dataset design (Wu et al., 2020; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Jaffe, 1986; Hall et al., 2005). The comparison to the published paper is intentionally narrow. The published paper argues that data analytics are most useful when firms innovate by recombining diverse existing knowledge and less useful for entirely new technologies. The local analogue translates that claim into patent-level AI scores, recombination breadth, and an entirely-new-technology indicator. The manuscript therefore evaluates concordance at the level of mechanism, sign pattern, and construct behavior before drawing any conclusion about exact numerical replication (Wu et al., 2020; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001). This comparison rule matters because the local paper is based only on processed datasets and verified outputs already present in the workspace. The present manuscript is classified as methodology-specific local analogue. When the original paper depends on proprietary, historical, or otherwise unavailable preprocessing layers, a full byte-identical recreation is not claimed. The more defensible question is whether the local evidence would lead a careful reader to a similar substantive interpretation (Wu et al., 2020; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and

Nerkar, 2001).

8 Discussion

The published paper argues that data analytics are most useful when firms innovate by recombining diverse existing knowledge and less useful for entirely new technologies. The local analogue translates that claim into patent-level AI scores, recombination breadth, and an entirely-new-technology indicator (Wu et al., 2020; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001). Taken together, the results are most informative about whether the source paper's mechanism is directionally recoverable in local data rather than whether every original coefficient can be duplicated exactly (Wu et al., 2020; Jaffe, 1986; Hall et al., 2005; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996). The first limitation concerns measurement. The published paper uses direct firm-level evidence on data analytics capabilities and combines survey and patent analyses (Wu et al., 2020; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Jaffe, 1986; Hall et al., 2005). The second limitation concerns sample translation. The local rerun uses patent-level AI classification scores as the executable analytics-capability analogue (Wu et al., 2020; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Jaffe, 1986; Hall et al., 2005). The third limitation concerns interpretation. The replication should therefore be read as a patent-data analogue of the complementarity logic rather than a direct replay of the original survey architecture (Wu et al., 2020; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Jaffe, 1986; Hall et al., 2005).

9 Conclusion

The paper's contribution is a transparent patent-level analogue that shows whether analytics-coded capability is more complementary to diverse recombination than to entirely new technological moves in the current local dataset (Wu et al., 2020; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and

Nerkar, 2001; Jaffe, 1986; Hall et al., 2005). The paper therefore contributes a traceable local replication record that can be rerun, audited, and extended as additional datasets or original-source details become available (Wu et al., 2020; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996; Jaffe, 1986; Hall et al., 2005).

A Appendix A: Supplementary Sample Reconstruction

The sample table below shows the observable route from the broader local file universe to the final estimating sample. It is included to make sample narrowing visible and to show where exclusions enter the paper.

Statistic	Value	RawStatistic
Observations	23,901	rows
Unique Firms	184	unique_firms
Start Year	1986	year_min
End Year	2016	year_max
Mean Impact	5.309	mean_impact
Share Ai	8.74%	share_ai
Mean Ai Score	0.136	mean_ai_score
Mean Diverse Recombination	4.291	mean_diverse_recombination
Share Entirely New	0.00%	share_entirely_new

Detailed sample reconstruction table.

B Appendix B: Supplementary Descriptive Tables

The descriptive appendix documents the scale, spread, and support of the variables used in the paper. These moments matter because they determine how the estimated coefficients should be read in substantive rather than merely statistical terms.

Variable	N	Mean	SD	P25	Median	P75	Min	Max
Inventive impact	23,901	5.309	17.53	1.000	2.000	6.000	0	812
AI patent indicator	23,901	0.0874	0.282	0.0000	0.0000	0.0000	0	1
AI score	23,901	0.136	0.294	0.0038	0.0096	0.0419	0	1
Breadth of recombination	23,901	4.291	4.563	2.000	3.000	5.000	2	117
Entirely new technology	23,901	0.0000	0.0000	0.0000	0.0000	0.0000	0	0
Science signal	23,901	6.772	2.130	5.729	7.149	8.163	0	10
Recombination degree	23,901	0.555	0.174	0.444	0.560	0.691	0	1
Same-year patents	23,901	727.1	1,302	109.0	311.0	652.0	1	7321

Descriptive statistics appendix.

C Appendix C: Correlation Matrix

Variable	Inventive impact	AI patent indicator	AI score	Breadth of recombination	Entirely new technology	Science signal	Recombination degree	Same-year patents
Inventive impact	1.000	0.026	0.027	0.364	NA	0.020	0.018	0
AI patent indicator	0.026	1.000	0.884	0.046	NA	0.041	0.066	0
AI score	0.027	0.884	1.000	0.055	NA	0.055	0.073	0
Breadth of recombination	0.364	0.046	0.055	1.000	NA	0.023	0.466	0
Entirely new technology	NA	NA	NA	NA	NA	NA	NA	NA
Science signal	0.020	0.041	0.055	0.023	NA	1.000	0.037	0
Recombination degree	0.018	0.066	0.073	0.466	NA	0.037	1.000	0
Same-year patents	-0.018	0.224	0.266	0.011	NA	0.463	0.065	1

Correlation matrix for the main numeric variables.

D Appendix D: Yearly Audit

The yearly audit provides a simple check on whether the local panel is driven by a narrow time slice or whether the variables and sample counts evolve over a broad observed window.

Period	Patents	MeanImpact	AIShare	MeanAIScore	MeanDiverseRecombination
1986	646	4.015	1.86%	0.0560	2.865
1987	793	4.296	1.89%	0.0594	2.914
1988	1,071	4.469	2.43%	0.0532	3.135
1989	1,398	3.926	2.43%	0.0573	3.084
1990	1,360	4.507	2.28%	0.0590	3.117
1991	1,333	4.749	2.18%	0.0503	3.151
1992	1,309	5.007	2.52%	0.0572	3.329
1993	1,358	5.678	4.05%	0.0799	3.465
1994	1,337	6.272	5.53%	0.0908	3.572
1995	1,217	6.260	5.26%	0.0960	3.844
1996	659	6.727	5.92%	0.102	3.838
1997	646	6.014	6.66%	0.114	4.245
1998	356	6.295	2.25%	0.0578	3.924
1999	654	5.483	5.05%	0.0875	3.810
2000	636	5.541	4.09%	0.0875	4.022
2001	614	5.151	5.86%	0.102	4.414
2002	632	4.889	6.17%	0.108	4.384
2003	522	4.502	5.17%	0.104	4.418
2004	324	3.664	6.79%	0.112	4.809
2005	284	4.500	14.08%	0.196	5.486
2006	336	4.890	13.99%	0.198	5.265
2007	246	3.654	8.13%	0.153	5.268
2008	535	3.652	27.10%	0.357	4.512
2009	206	4.801	15.53%	0.223	6.005
2010	511	4.417	11.35%	0.165	6.648
2011	651	6.452	16.90%	0.232	6.275
2012	1,068	4.000	21.07%	0.290	6.519
2013	868	5.386	16.59%	0.247	6.730

Period	Patents	MeanImpact	AIShare	MeanAIScore	MeanDiverseRecombination
2014	725	12.64	16.14%	0.240	6.989
2015	1,072	6.616	26.59%	0.343	6.174
2016	534	3.768	41.20%	0.511	4.629

Year-by-year sample audit.

E Appendix E: Grouped Outcome Patterns

Grouped summaries do not replace the formal model, but they show the raw outcome structure that the specification is attempting to explain. For that reason they are retained here as descriptive support rather than promoted to the status of a main estimate.

Quantile	N	MeanRegressor	MeanOutcome
Q1	4,781	0.0017	5.435
Q2	4,780	0.0047	4.894
Q3	4,780	0.0100	4.926
Q4	4,780	0.0333	5.109
Q5	4,780	0.632	6.183

Quantile summary for the main regressor.

F Appendix F: Full Coefficient Table and Fit Statistics

The full coefficient export is included for completeness so that readers can see the focal terms in the context of the broader specification. Where a fit file is available, it is appended here rather than in the main text to keep the article narrative readable.

term	estimate	std_error	statistic	p_value
ai_score_max_8	0.2098	0.0650	3.227	0.003
diverse_recombination	0.0419	0.0091	4.591	¡ .001
science_signal	0.0018	0.0289	0.0606	0.95
recombination_degree	-0.4400	0.0871	-5.054	¡ .001
total_patents_same_year	-4.48e-05	2.08e-05	-2.157	0.039
total_patents_until_prev_log	-0.1230	0.0648	-1.897	0.067
ai_score_max_8:diverse_recombination	-0.0024	0.0072	-0.331	0.74

Full coefficient dump exported from the local estimation script.

G Appendix G: Coverage Tables

Variable	Missing	MissingShare
Inventive impact	0	0.00%
AI patent indicator	0	0.00%
AI score	0	0.00%
Breadth of recombination	0	0.00%
Entirely new technology	0	0.00%
Science signal	0	0.00%
Recombination degree	0	0.00%
Same-year patents	0	0.00%

Missingness audit for the variables summarized in the manuscript.

Entity	Label	Observations	FirstPeriod	LastPeriod
6066	INTERNATIONAL BUSINESS MACHS COR	3,200	1986	2016
5047	GENERAL ELECTRIC CO	1,318	1986	2016
4194	EASTMAN KODAK CO	1,017	1986	2010
7435	MINNESOTA MINING & MFG CO	854	1986	2016
4087	DU PONT E I DE NEMOURS & CO	830	1986	2016
4839	FORD MOTOR CO DEL	815	1986	2016
7585	MOTOROLA INC	774	1986	2016
10499	TEXAS INSTRUMENTS INC	755	1986	2016
5073	GENERAL MOTORS CORP	671	1986	2008

Entity	Label	Observations	FirstPeriod	LastPeriod
4060	DOW CHEMICAL CO	574	1986	2016
1300	ALLIED SIGNAL INC	551	1986	2016
8762	PROCTER & GAMBLE CO	524	1986	2016
1690	APPLE COMPUTER INC	452	1990	2016
1581	AMERICAN TELEPHONE & TELEG CO	421	1986	2004
10983	UNITED TECHNOLOGIES CORP	403	1986	2016
12141	MICROSOFT CORP	398	1995	2016
4503	EXXON CORP	344	1986	2016
6266	JOHNSON & JOHNSON	340	1986	2016
2285	BOEING CO	334	1986	2016
7475	MOBIL CORP	300	1986	1998

Top entities by local observation count.

References

Ahuja, G. (2000). Collaboration networks, structural holes, and innovation: A longitudinal study. *Administrative Science Quarterly*, 45(3):425–455.

Ahuja, G. and Katila, R. (2001). Technological acquisitions and the innovation performance of acquiring firms: A longitudinal study. *Strategic Management Journal*, 22(3):197–220.

Ahuja, G. and Lampert, C. M. (2001). Entrepreneurship in the large corporation: A longitudinal

study of how established firms create breakthrough inventions. *Strategic Management Journal*, 22(6-7):521–543.

Almeida, P. and Kogut, B. (1999). Localization of knowledge and the mobility of engineers in regional networks. *Management Science*, 45(7):905–917.

Anderson, P. and Tushman, M. L. (1986). Technological discontinuities and organizational environments. *Administrative Science Quarterly*, 31(3):439–465.

Benner, M. J. and Tushman, M. L. (2003). Exploitation, exploration, and process management: The productivity dilemma revisited. *Academy of Management Review*, 28(2):238–256.

Cohen, W. M. and Levinthal, D. A. (1990). Absorptive capacity: A new perspective on learning and innovation. *Administrative Science Quarterly*, 35(1):128–152.

Fleming, L. (2001). Recombinant uncertainty in technological search. *Management Science*, 47(1):117–132.

Fleming, L., Mingo, S., and Chen, D. (2007). Collaborative brokerage, generative creativity, and creative success. *Administrative Science Quarterly*, 52(3):443–475.

Fleming, L. and Sorenson, O. (2004). Science as a map in technological search. *Strategic Management Journal*, 25(8-9):909–928.

Ghosh, A., Martin, X., Pennings, J. M., and Wezel, F. C. (2014). Ambition is nothing without focus: Compensating for negative transfer of experience in r&d. *Organization Science*, 25(2):572–590.

Hall, B. H., Jaffe, A., and Trajtenberg, M. (2005). Market value and patent citations. *RAND Journal of Economics*, 36(1):16–38.

Hargadon, A. and Sutton, R. I. (1997). Technology brokering and innovation in a product development firm. *Administrative Science Quarterly*, 42(4):716–749.

Henderson, R. and Cockburn, I. (1994). Measuring competence? exploring firm effects in pharmaceutical research. *RAND Journal of Economics*, 25(1):63–84.

Henderson, R. M. and Clark, K. B. (1990). Architectural innovation: The reconfiguration of existing product technologies and the failure of established firms. *Administrative Science Quarterly*, 35(1):9–30.

Jaffe, A. B. (1986). Technological opportunity and spillovers of r&d: Evidence from firms' patents, profits, and market value. *American Economic Review*, 76(5):984–1001.

Jaffe, A. B., Trajtenberg, M., and Henderson, R. (1993). Geographic localization of knowledge spillovers as evidenced by patent citations. *Quarterly Journal of Economics*, 108(3):577–598.

Katila, R. and Ahuja, G. (2002). Something old, something new: A longitudinal study of search behavior and new product introduction. *Academy of Management Journal*, 45(6):1183–1194.

Kogut, B. and Zander, U. (1992). Knowledge of the firm, combinative capabilities, and the replication of technology. *Organization Science*, 3(3):383–397.

Kogut, B. and Zander, U. (1996). What firms do? coordination, identity, and learning. *Organization Science*, 7(5):502–518.

Lavie, D., Kang, J., and Rosenkopf, L. (2011). Balance within and across domains: The performance implications of exploration and exploitation in alliances. *Organization Science*, 22(6):1517–1538.

Levinthal, D. A. and March, J. G. (1993). The myopia of learning. *Strategic Management Journal*, 14(S2):95–112.

Levitt, B. and March, J. G. (1988). Organizational learning. *Annual Review of Sociology*, 14:319–338.

March, J. G. (1991). Exploration and exploitation in organizational learning. *Organization Science*, 2(1):71–87.

Nerkar, A. (2003). Old is gold? the value of temporal exploration in the creation of new knowledge. *Management Science*, 49(2):211–229.

Rivkin, J. W. and Siggelkow, N. (2003). Balancing search and stability: Interdependencies among elements of organizational design. *Management Science*, 49(3):290–311.

Rosenkopf, L. and Almeida, P. (2003). Overcoming local search through alliances and mobility. *Management Science*, 49(6):751–766.

Rosenkopf, L. and Nerkar, A. (2001). Beyond local search: Boundary-spanning, exploration, and impact in the optical disk industry. *Strategic Management Journal*, 22(4):287–306.

Schilling, M. A. and Phelps, C. C. (2007). Interfirm collaboration networks and knowledge creation: The impact of large-scale network structure on firm innovation. *Management Science*, 53(7):1113–1126.

Stuart, T. E. (2000). Interorganizational alliances and the performance of firms: A study of growth and innovation rates in a high-technology industry. *Strategic Management Journal*, 21(8):791–811.

Stuart, T. E. and Podolny, J. M. (1996). Local search and the evolution of technological capabilities. *Strategic Management Journal*, 17(S1):21–38.

Teece, D. J., Pisano, G., and Shuen, A. (1997). Dynamic capabilities and strategic management. *Strategic Management Journal*, 18(7):509–533.

Trajtenberg, M. (1990). A penny for your quotes: Patent citations and the value of innovations. *RAND Journal of Economics*, 21(1):172–187.

Tripsas, M. and Gavetti, G. (2000). Capabilities, cognition, and inertia: Evidence from digital imaging. *Strategic Management Journal*, 21(10-11):1147–1161.

Wu, L., Hitt, L., and Lou, B. (2020). Data analytics, innovation, and firm productivity. *Management Science*, 66(5):2017–2039.